

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. Source-dependent microcanonical preparation is proved by a full complex-pressure Fourier theorem, not by spatially independent dynamical blocks. The singleton compound-Poisson pressure and the normally convergent trajectory-polymer remainder give Gaussian, compact-annulus, and speed-dependent high-frequency gaps. A degree-one finite saddle and a relative local theorem for regular CLT shells yield the constrained pressure and covariance Schur complement.

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1. ROUND-TWELVE POSITIVE CLOSURE: POLYMER CHARACTERISTIC PRESSURE AND SHARP REGULAR SHELLS

The proof uses the full complex pressure of the dynamically tilted gas. It does not partition the fixed torus into linearly many causally separated blocks. Compact and high frequencies are controlled directly by the singleton term and the absolutely convergent real-trajectory polymer remainder.

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1.1. Pressure decomposition and strict complex aperiodicity. For a path/contact source H and constraint multiplier λ write

$$Q_\varepsilon(H, \lambda) = \frac{1}{\mu_\varepsilon} \log \mathbb{E} \exp\{\mu_\varepsilon H + \lambda \cdot \sum_i C(z_i(0))\}.$$

The B2 grand-canonical expansion gives

$$Q_\varepsilon = Q_\varepsilon^{(1)} + R_\varepsilon, \quad \sup_{\mathcal{U}} \|D^j R_\varepsilon\| \leq C_j T_*, \quad j \leq 4,$$

on a complex source-multiplier polydisc. Here $Q^{(1)}$ is the exact compound-Poisson singleton pressure and T_* is chosen below.

Lemma 1.1 (Uniform constraint covariance). *Assume that the mark map C has full affine span on one finite family of positive-activity phase boxes. There are $T_* > 0$, $c > 0$, and a common polydisc such that*

$$D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda) \geq cI$$

for every real (H, λ) in the polydisc and all small ε .

Proof. The singleton Hessian is the activity integral of $(C - \bar{C})(C - \bar{C})^T$ and is uniformly positive by the affine-span hypothesis and positivity on the chosen boxes. The connected Hessian is bounded by CT_* by the B2 trajectory-cluster norm. Choose T_* so that this perturbation is at most half the singleton lower eigenvalue. $\square$

Let $(t, u) \in [-\pi, \pi] \times \mathbb{R}^{d-1}$ be the number-lattice and continuous frequencies. At the exact finite saddle set

$$\Phi_\varepsilon(t, u) = Q_\varepsilon(H, \lambda_{H,a,\varepsilon} + i(t, u)) - Q_\varepsilon(H, \lambda_{H,a,\varepsilon}).$$

Theorem 1.2 (Full-frequency pressure gap). *There are $\delta, R, c > 0$ and, for every $s > d + 3$, a C_s such that*

$$\Re \Phi_\varepsilon(t, u) \leq \begin{cases} -c(|t|^2 + |u|^2), & |t| + |u| \leq \delta, \\ -c, & \delta \leq |t| + |u| \leq R, \\ -c - C_s \log(1 + |u|), & |u| > R, \end{cases}$$

uniformly on compact source and target charts. Equivalently the full characteristic function is Gaussian near zero, exponentially small in μ_ε on every compact nonzero annulus, and bounded by

$$e^{-c\mu_\varepsilon} (1 + |u|)^{-s\mu_\varepsilon}$$

in the continuous tail. The three regions cover the complete Fourier fundamental domain.

Proof. Near zero use Lemma 1.1 and Taylor expansion. On a compact annulus, equality in the triangle inequality for the tilted singleton integral would make $(t, u) \cdot C$ constant modulo 2π on every full-rank phase box. The number span and continuous affine span force the trivial frequency. Compactness gives a negative singleton gap; the $O(T_*)$ polymer remainder is

absorbed by the choice of T_* . For large u , integrate the smooth singleton activity on the full-rank boxes s times. Its Fourier transform is $O((1+|u|)^{-s})$, while the Poisson exponent contains μ_ε independent singleton opportunities at the logarithmic pressure level. Normal convergence of connected polymers changes the real part by at most CT_* and does not undo the negative activity mass. This gives the final speed-dependent power. $\square$

1.2. The finite saddle and its mean image.

Theorem 1.3 (Uniform finite-volume mean map). *There is a bounded convex multiplier domain Λ and a compact target set A in the relative interior of the constraint convex support such that, for every H in the source chart and small ε ,*

$$\lambda \mapsto D_\lambda Q_\varepsilon(H, \lambda)$$

is a degree-one diffeomorphism from Λ onto a neighborhood of A . The exact saddle $\lambda_{H,a,\varepsilon}$ and all derivatives up to order three converge locally uniformly to their limiting values.

Proof. Strict convexity gives local injectivity. Choose Λ as a smooth sublevel of the singleton Legendre function containing the limiting saddles for A . On its boundary the singleton gradient points strictly outward relative to A ; the connected pressure is a CT_* perturbation, so the same holds for the full finite pressure. Brouwer degree is one and no two preimages can occur by strict convexity. Normal complex convergence and the implicit-function theorem give derivative convergence. $\square$

1.3. Regular shell coefficients. Let $W_\varepsilon \subset \mathbb{R}^{d-1}$ be centered Lipschitz domains in CLT coordinates satisfying

$$\gamma_\Sigma(W_\varepsilon) \geq \mu_\varepsilon^{-M}, \quad \gamma_\Sigma((\partial W_\varepsilon)^{+h}) \leq Ch\mu_\varepsilon^M$$

for $0 < h < 1$, and let the number coordinate be exact. Centered expanding boxes with $\sqrt{\mu}\delta_\varepsilon \rightarrow \infty$ are included.

Theorem 1.4 (Relative mixed local coefficient). *Uniformly on the source and target chart,*

$$\begin{aligned} \mathbb{P}_{H,\lambda_{H,a,\varepsilon}}(N = N_\varepsilon, \sqrt{\mu_\varepsilon}(Y - a) \in W_\varepsilon) \\ = \frac{\gamma_{\Sigma_{H,a}^c}(W_\varepsilon)}{\sqrt{2\pi\mu_\varepsilon\sigma_{N|Y}^2}} (1 + o(1)), \end{aligned}$$

where the error is relative to the displayed Gaussian mass. In particular its logarithm is $o(\mu_\varepsilon)$.

Proof. Fourier inversion is centered at the exact saddle. On a ball of radius $\mu^{-2/5}$, the fourth-order pressure bound gives a relative Edgeworth error. Smooth upper and lower approximations of 1_{W_ε} use a boundary thickness $h = \mu^{-K}$; the declared Minkowski estimate makes their Gaussian difference

$o(\gamma(W_\varepsilon))$. The compact annulus and tail are $e^{-c\mu}$ times an integrable speed-dependent power by Theorem 1.2, hence are $o(\mu^{-1/2}\gamma(W_\varepsilon))$. Division by the Gaussian main term gives the relative formula. $\square$

1.4. Source-dependent microcanonical transfer.

Theorem 1.5 (Prepared pressure and initial covariance). *For the primitive shell in Theorem 1.4,*

$$Q^{\text{mc},a}(H) = Q(H, \lambda_{H,a}) - \lambda_{H,a} \cdot a - Q(0, \lambda_{0,a}) + \lambda_{0,a} \cdot a.$$

The initial empirical law has the constrained entropy rate, and its Gaussian covariance is

$$\Sigma^{\text{mc}} = \Sigma - \Sigma C^* (C \Sigma C^*)^{-1} C \Sigma.$$

The theorem applies jointly to the B2 particle-path and actual-contact source class.

Proof. Change measure in numerator and denominator at their separate exact finite saddles. Theorem 1.4 shows that both residual shell probabilities have subexponential logarithm and supplies their relative coefficients. Passing to the normal pressure limit gives the displayed formula. Convex duality gives the initial rate. Differentiating the saddle equation twice gives the Schur complement covariance. The only dynamical input is the earlier B2 grand-canonical pressure theorem, so the dependency is noncircular. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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